Label-efficient Land Cover Classification at High Spatial Resolution using Multi-stage Pre-training

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Land Use and Land Cover (LULC)

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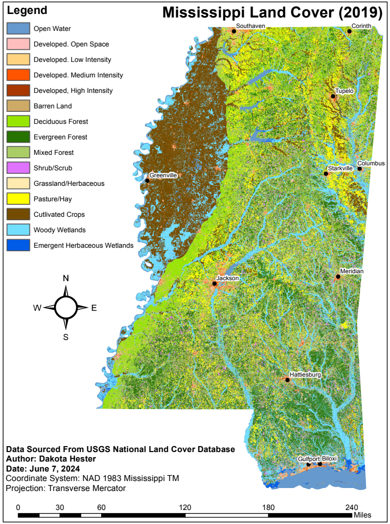

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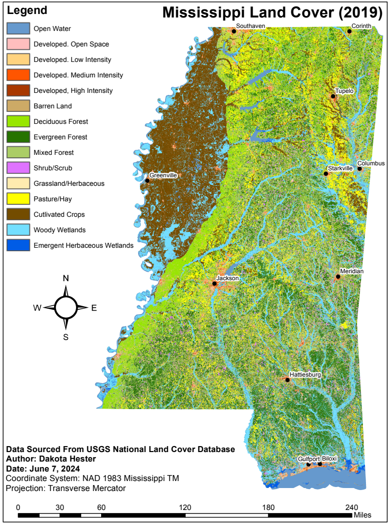

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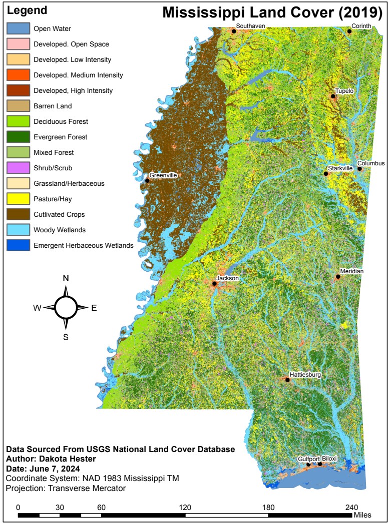
- LULC data are used in a wide range of disciplines, including agriculture, forestry, urban planning, and environmental science [4–6]



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  - Ancillary information (i.e., topography, soil, etc) [6, 7]
- Non-parametric, supervised classifiers are often used (e.g., Decision Trees, Random Forests, Support Vector Machines, etc.) [8, 9].

Pixel-wise Classification

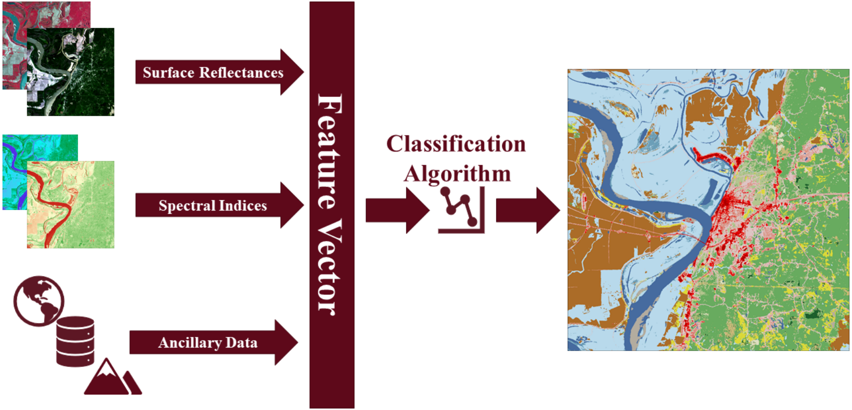

Figure 2: Pixel-wise classification workflow.

Land Cover at High Spatial Resolution

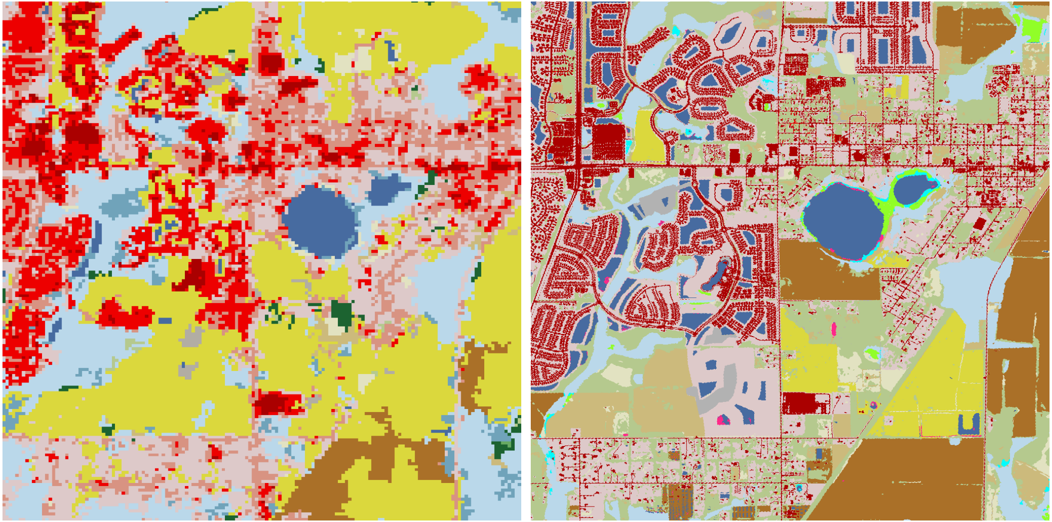

Figure 3: Comparison of land cover maps at 30 m (left) and 1 m (right) spatial resolution.

Decreased Class Discriminability

Classifying pixels gets more difficult as spatial resolution increases [10, 11].

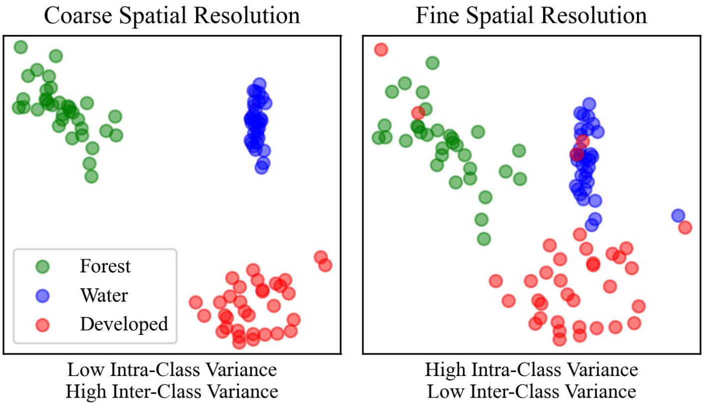

Figure 4: As spatial resolution increases, inter-class variance decreases [11].

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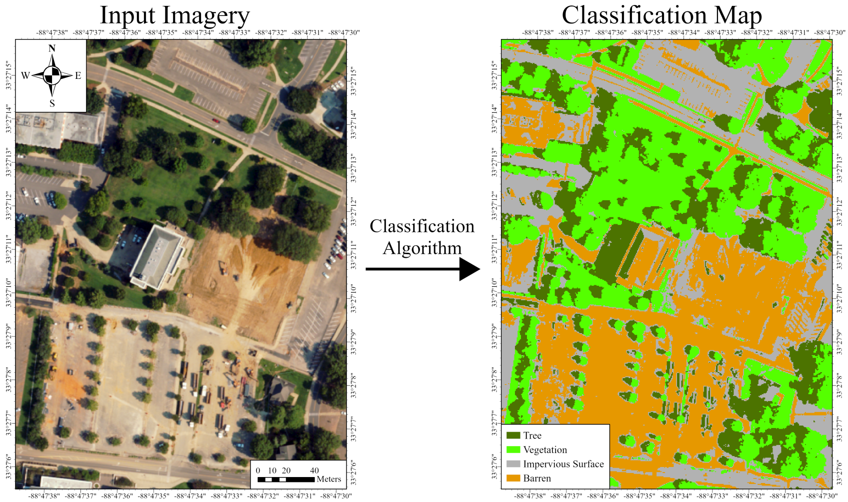

Figure 5: Applying a pixel-wise classifier to high spatial resolution data typically results in noisy classification maps.

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  3. End-to-end learning — unlike traditional methods, deep models can be trained in an end-to-end fashion, where little to no pre-processing or post-processing is required [14–17].

Convolutional Neural Networks (CNNs)

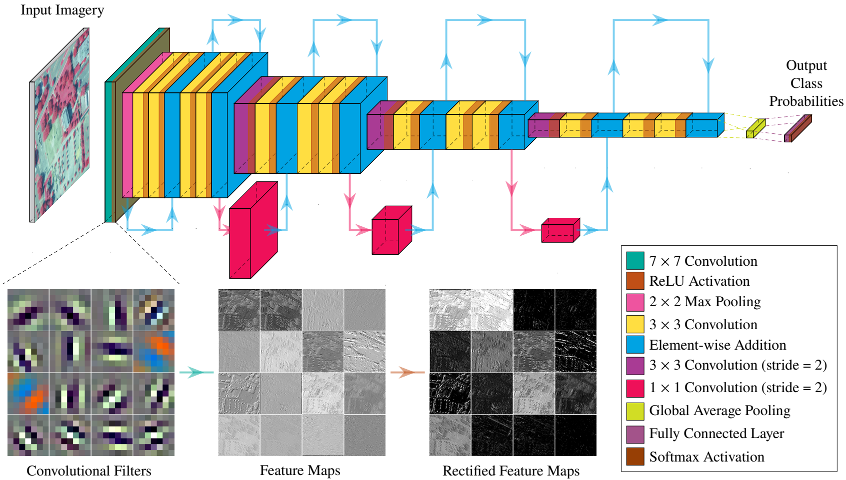

Figure 6: A ResNet-18 CNN architecture [18]. A sampling of learned convolutional filters from the first layer is shown.

U-Net

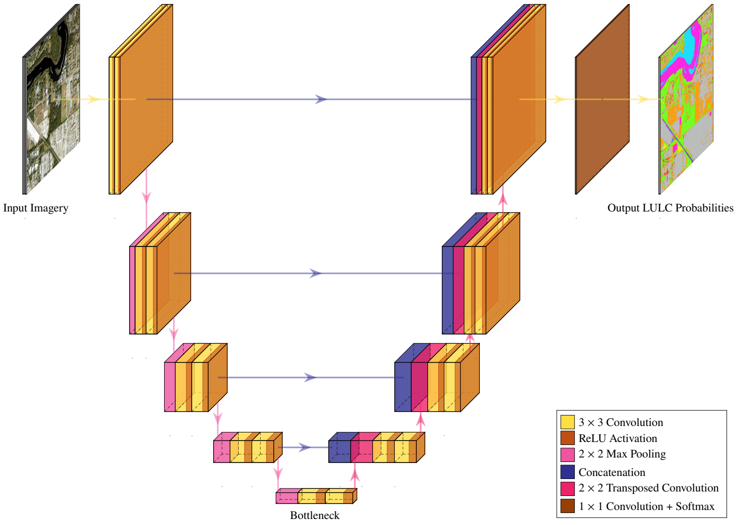

Figure 7: U-Net architecture [19].

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- For land cover, this is especially problematic, as high resolution data is complex and difficult to annotate at scale [20].
- One approach to mitigate this issue is to use transfer learning, where a network can be pre-trained on a related task with a large amount of data, then fine-tuned on the target task with a smaller amount of data [21].

Transfer Learning

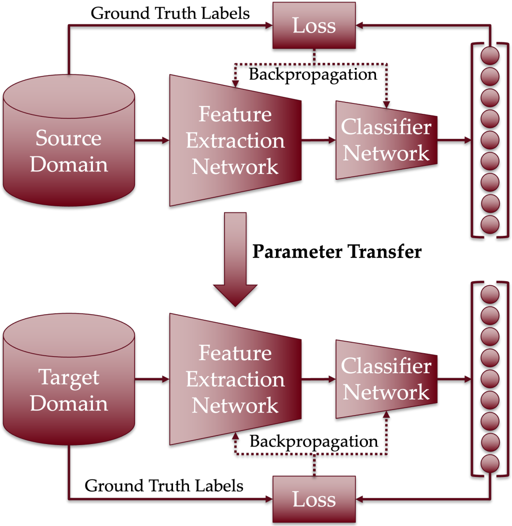

Figure 8: Transfer learning paradigm.

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- In effect, noise is added to an input image, then train the network to predict the original image from the noisy image.
- The network needs to learn features that are robust to noise while being meaningful for reconstruction [24].

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- We construct a large U-Net model that undergoes various pre-training regimens, then fine-tune the model on only 119 tiles of fully annotated data from the state of Mississippi.
- Transfer learning is not an uncommon approach, but self-supervised pre-training is not common in the remote sensing community — especially in the context of land cover classification. As such, we are most interested in the performance of our self-supervised model.

Training Strategy

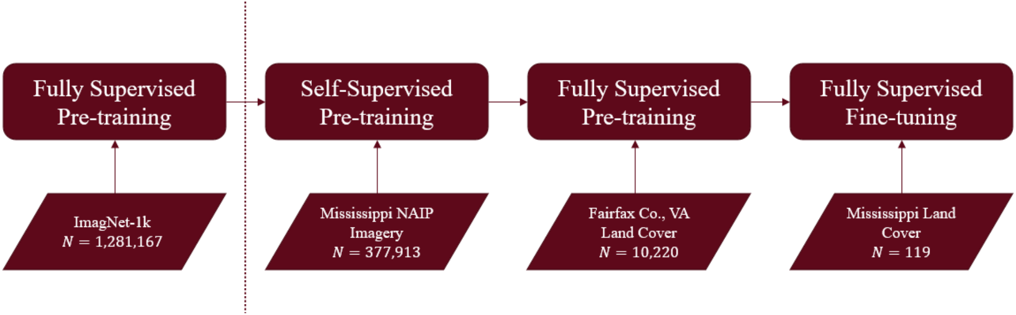

Figure 9: Model training flowchart.

Pre-training Strategies

Pre-training with a denoising autoencoder improved the performance of the model under the 2-stage pre-training strategy.

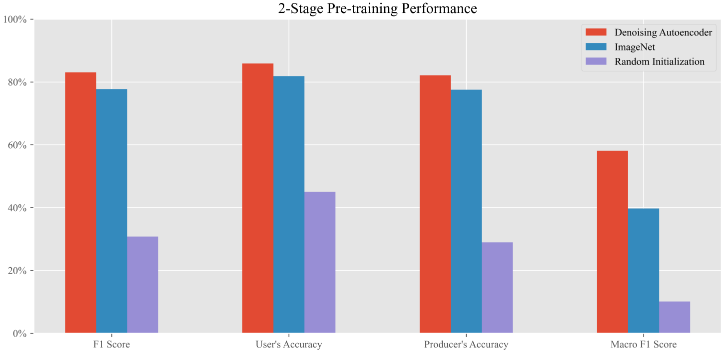

Figure 10: Performance of models on the Mississippi land cover dataset.

Pre-training Strategies

An additional fully supervised pre-training stage substantially improves the performance of the model.

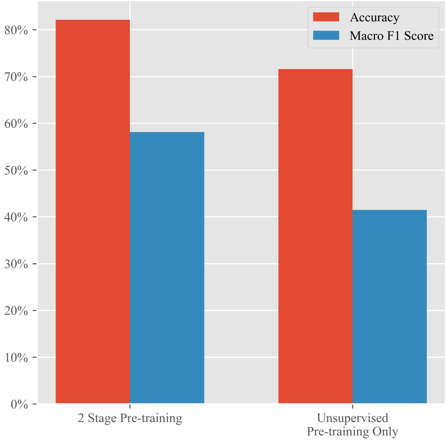

Figure 11: Performance of the best model with and without additional supervised pre-training.

Confusion Matrix

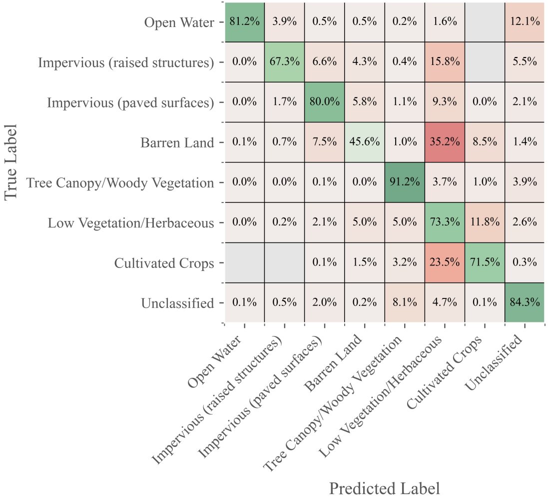

Figure 12: Confusion matrix of the final model on the Mississippi land cover dataset.

Mississippi Hi-Res Land Cover

Figure 13: 2023 Mississippi Land Cover Map.

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Summary

- Large scale pre-training of a deep model using an unsupervised regimen can enable the training of an over-parameterized deep model on a small amount of labeled data.
- Further, additional pre-training using a modest amount of labeled data can boost the performance of the model even further.
- Still, classifying more visually intricate land cover classes remains a challenge, even with a large model.

Future Works

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- Post-processing using OBIA-inspired techniques to refine classification map edges.
- Applying model to super-resolved satellite imagery.

Acknowledgements

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Mississippi Center for Supercomputing Research

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